B ADDITIONAL EXPERIMENTAL DETAILS

B.1 ALTERNATIVE ATTENTION AGREEMENT METRIC

Here we present an alternative formulation to Eq. 1 based on an attention-weighted average. We define an indicator function $f(i, j)$ for property f that returns 1 if the property is present in token pair (i, j) (i.e., if amino acids i and j are in contact), and zero otherwise. We then compute the proportion of attention that matches with f over a dataset $\boldsymbol{X}$ as follows:

$$p_\alpha(f) = \sum_{x \in X} \sum_{i=1}^{|x|} \sum_{j=1}^{|x|} f(i,j)\alpha_{i,j}(x) \Big/ \sum_{x \in X} \sum_{i=1}^{|x|} \sum_{j=1}^{|x|} \alpha_{i,j}(x) \tag{3}$$

where $\alpha_{i,j}(x)$ denotes the attention from i to j for input sequence x.

B.2 STATISTICAL SIGNIFICANCE TESTING AND NULL MODELS

We perform statistical significance tests to determine whether any results based on the metric defined in Equation 1 are due to chance. Given a property f, as defined in Section 3, we perform a two-proportion z-test comparing (1) the proportion of high-confidence attention arcs ($\alpha_{i,j} > \theta$) for which $f(i, j) = 1$, and (2) the proportion of all possible pairs i, j for which $f(i, j) = 1$. Note that the first proportion is exactly the metric $p_\alpha(f)$ defined in Equation 1 (e.g. the proportion of attention aligned with contact maps). The second proportion is simply the background frequency of the property (e.g. the background frequency of contacts). Since we extract the maximum scores over all of the heads in the model, we treat this as a case of multiple hypothesis testing and apply the Bonferroni correction, with the number of hypotheses m equal to the number of attention heads.

As an additional check that the results did not occur by chance, we also report results on baseline (null) models. We initially considered using two forms of null models: (1) a model with randomly initialized weights. and (2) a model trained on randomly shuffled sequences. However, in both cases, none of the sequences in the dataset yielded attention weights greater than the attention threshold θ. This suggests that the mere existence of the high-confidence attention weights used in the analysis could not have occurred by chance, but it does not shed light on the particular analyses performed. Therefore, we implemented an alternative randomization scheme in which we randomly shuffle attention weights from the original models as a post-processing step. Specifically, we permute the sequence of attention weights *from* each token for every attention head. To illustrate, let's say that the original model produced attention weights of (0.3, 0.2, 0.1, 0.4, 0.0) from position i in protein sequence x from head h, where $|x| = 5$. In the null model, the attention weights from position i in sequence x in head h would be a random permutation of those weights, e.g., (0.2, 0.0, 0.4, 0.3, 0.1). Note that these are still valid attention weights as they would sum to 1 (since the original weights would sum to 1 by definition). We report results using this form of baseline model.

B.3 PROBING METHODOLOGY

Embedding probe. We probe the embedding vectors output from each layer using a linear probing classifier. For token-level probing tasks (binding sites, secondary structure) we feed each token's output vector directly to the classifier. For token-pair probing tasks (contact map) we construct a pairwise feature vector by concatenating the elementwise differences and products of the two tokens' output vectors, following the TAPE[4] implementation.

We use task-specific evaluation metrics for the probing classifier: for secondary structure prediction, we measure F1 score; for contact prediction, we measure precision@$L/5$, where L is the length of the protein sequence, following standard practice (Moult et al., 2018); for binding site prediction, we measure precision@$L/20$, since approximately one in twenty amino acids in each sequence is a binding site (4.8% in the dataset).

Attention probe. Just as the attention weight $\alpha_{i,j}$ is defined for a pair of amino acids (i, j), so is the contact property $f(i, j)$, which returns true if amino acids i and j are in contact. Treating the attention weight as a feature of a token-pair (i, j), we can train a probing classifier that predicts the

[4] `https://github.com/songlab-cal/tape`

contact property based on this feature, thereby quantifying the attention mechanism's knowledge of that property. In our multi-head setting, we treat the attention weights across all heads in a given layer as a feature vector, and use a probing classifier to assess the knowledge of a given property in the attention weights across the entire layer. As with the embedding probe, we measure performance of the probing classifier using precision@$L/5$, where L is the length of the protein sequence, following standard practice for contact prediction.

B.4 Datasets

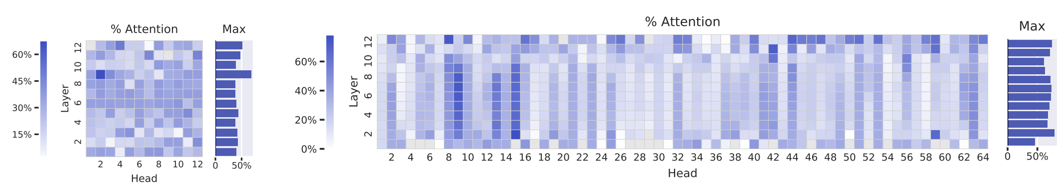
We used two protein sequence datasets from the TAPE repository for the analysis: the ProteinNet dataset (AlQuraishi, 2019; Fox et al., 2013; Berman et al., 2000; Moult et al., 2018) and the Secondary Structure dataset (Rao et al., 2019; Berman et al., 2000; Moult et al., 2018; Klausen et al., 2019). The former was used for analysis of amino acids and contact maps, and the latter was used for analysis of secondary structure. We additionally created a third dataset for binding site and post-translational modification (PTM) analysis from the Secondary Structure dataset, which was augmented with binding site and PTM annotations obtained from the Protein Data Bank's Web API.[5] We excluded any sequences for which annotations were not available. The resulting dataset sizes are shown in Table 2. For the analysis of attention, a random subset of 5000 sequences from the training split of each dataset was used, as the analysis was purely evaluative. For training and evaluating the diagnostic classifier, the full training and validation splits were used.

Table 2: Datasets used in analysis

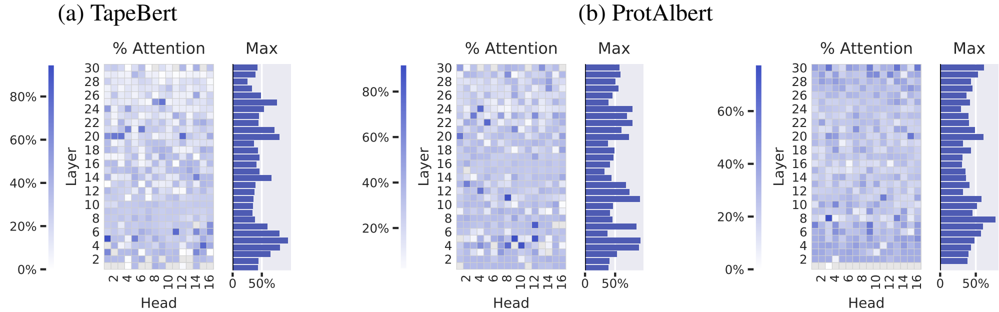
Dataset	Train size	Validation size
ProteinNet	25299	224
Secondary Structure	8678	2170
Binding Sites / PTM	5734	1418

C Additional Results of Attention Analysis

C.1 Secondary Structure



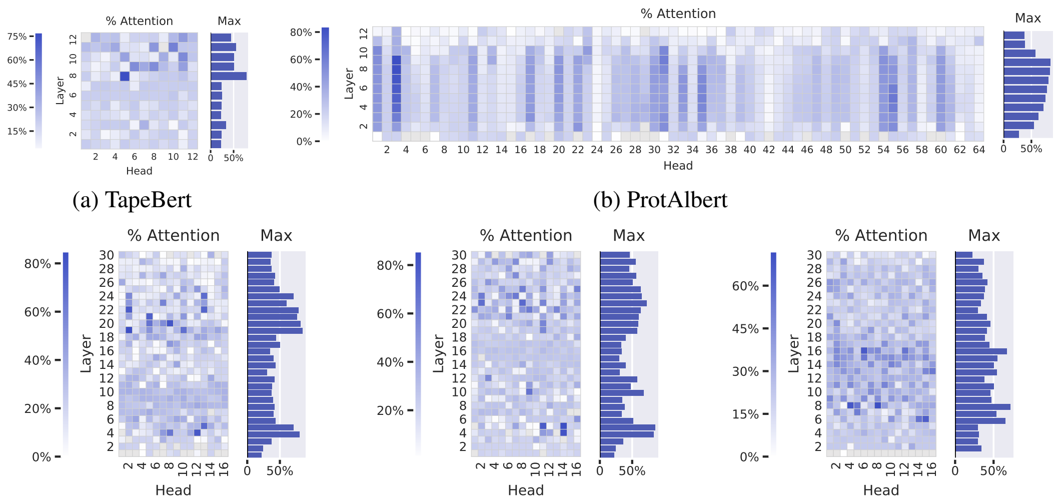
(a) TapeBert (b) ProtAlbert



(c) ProtBert (d) ProtBert-BFD (e) ProtXLNet

Figure 8: Percentage of each head's attention that is focused on *Helix* secondary structure.

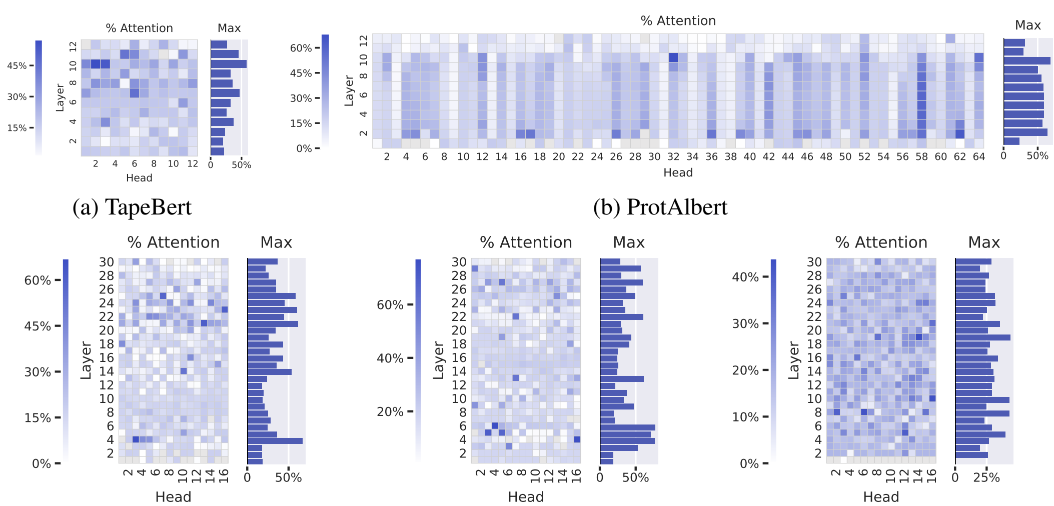
[5]`http://www.rcsb.org/pdb/software/rest.do`



(a) TapeBert (b) ProtAlbert

(c) ProtBert (d) ProtBert-BFD (e) ProtXLNet

Figure 9: Percentage of each head's attention that is focused on *Strand* secondary structure.



(a) TapeBert (b) ProtAlbert

(c) ProtBert (d) ProtBert-BFD (e) ProtXLNet

Figure 10: Percentage of each head's attention that is focused on *Turn/Bend* secondary structure.